

# Introduction — The Problem of Measure

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Consider the set of all real numbers,  $\mathbb{R}$ . **Our goal is to assign a *measure* to every subset of  $\mathbb{R}$ .** Without defining this term rigorously, we have a certain intuitive notion of how this measure on  $\mathbb{R}$  corresponds to length. For instance, it makes sense to say that the measure of the interval  $(a, b)$  is  $b - a$ , and that translating this interval by a certain quantity, say  $x$ , should not result in a change in its measure; that is, the measure of  $(a + x, b + x)$  should remain  $b - a$  for all  $x \in \mathbb{R}$ . We would like to extend this notion from open intervals like  $(a, b)$  to all subsets of  $\mathbb{R}$ .

Our objective, then, is to find a function  $\lambda$  that satisfies the following:

(P1) To every subset of  $\mathbb{R}$ ,  $\lambda$  should assign some nonnegative value, possibly  $+\infty$ . Hence, it should be a function from  $\mathcal{P}(\mathbb{R})$  to  $[0, +\infty]$ , i.e.,  $\lambda : \mathcal{P}(\mathbb{R}) \rightarrow [0, +\infty]$ .

(P2)  $\lambda((a, b)) = b - a$ .

(P3)  $\lambda(A + x) = \lambda(A)$ , where  $A + x := \{a + x : a \in A\}$ .

(P4) Let  $A_1, A_2, \dots$  be a sequence of pairwise disjoint subsets of  $\mathbb{R}$ . Then,  $\lambda(\bigsqcup_{j \geq 1} A_j) = \sum_{j \geq 1} \lambda(A_j)$ .

**We now show that such a function cannot exist.**

**Lemma.** Let  $E$  and  $F$  be subsets of  $\mathbb{R}$  such that  $E \subseteq F$ . Then,  $\lambda(E) \leq \lambda(F)$ .

*Proof.* We first show that  $\lambda(\emptyset) = 0$ . Set  $A_j = \emptyset$  for all  $j \geq 1$  in (P4). Then,  $\lambda(\emptyset) = \lambda(\bigsqcup_{j \geq 1} \emptyset) = \sum_{j \geq 1} \lambda(\emptyset)$ , which is possible only if  $\lambda(\emptyset) = 0$ .

If  $E \subseteq F$ , then  $F = E \cup (F \setminus E)$ , and  $E$  and  $F \setminus E$  are disjoint. Setting  $A_1 = E$ ,  $A_2 = F \setminus E$ , and  $A_j = \emptyset$  for all  $j \geq 3$  in (P4), we get  $\lambda(F) = \lambda(E) + \lambda(F \setminus E) \geq \lambda(E)$ . **QED**

**Theorem.** A function  $\lambda$  satisfying (P1) to (P4) does not exist.

*Proof.* Suppose, for contradiction, that such a function  $\lambda$  does exist.

Define a relation  $\sim$  such that  $a \sim b$  if and only if  $a - b \in \mathbb{Q}$ .

Clearly,  $\sim$  is an equivalence relation. Define  $[x] := \{y \in \mathbb{R} : x - y \in \mathbb{Q}\}$ . Then,  $[x]$  is an equivalence class. Observe that  $[x] = \{x + q : q \in \mathbb{Q}\}$ . Hence, there is a bijection  $q \mapsto x + q$  from  $\mathbb{Q}$  to  $[x]$ , and so  $[x]$  is a countable set.

Let  $\Lambda$  be the collection of all equivalence classes. Then,  $\Lambda$  is uncountably infinite. For if  $\Lambda$  were to be countably infinite, then, since each equivalence class is also countable, the union of all the equivalence classes would be a countable union of countable sets, which would

also be a countable set. However, the union of all equivalence classes is  $\mathbb{R}$ , since every real number belongs to some equivalence class, and  $\mathbb{R}$  is uncountable, which is a contradiction.

By the Axiom of Choice, we can choose one element from each equivalence class. Call the resulting set  $\Omega$ , and we may assume that  $\Omega \subseteq (0, 1)$ , for we can choose an element from each equivalence class that is in  $(0, 1)$ .

**We first show that  $\lambda(\Omega) = 0$ .**

**Let  $q_1, q_2 \in \mathbb{Q}$ ,  $q_1 \neq q_2$ . We now show that  $(\Omega + q_1) \cap (\Omega + q_2) = \emptyset$ .**

Let  $x \in (\Omega + q_1) \cap (\Omega + q_2)$ . Then,  $x \in \Omega + q_1$  and  $x \in \Omega + q_2$ , that is,  $x = \alpha + q_1$  and  $x = \beta + q_2$  for some  $\alpha, \beta \in \Omega$ . Hence,  $\alpha + q_1 = \beta + q_2$ , or,  $\alpha - \beta = q_2 - q_1 \in \mathbb{Q}$ . Therefore,  $\alpha \sim \beta$ , and so  $\alpha$  and  $\beta$  belong to the same equivalence classes. However, both  $\alpha$  and  $\beta$  are in  $\Omega$ , which contains exactly one element from each equivalence class, by construction. This implies that  $\alpha = \beta$ , and so  $q_1 = q_2$ , which contradicts that  $q_1 \neq q_2$ , proving that  $(\Omega + q_1) \cap (\Omega + q_2) = \emptyset$ .

**Now consider  $-1 < q < 1$ ,  $q \in \mathbb{Q}$ .** Since  $\Omega \subseteq (0, 1)$ ,

$$\bigsqcup_{\substack{-1 < q < 1 \\ q \in \mathbb{Q}}} (\Omega + q) \subseteq (-1, 2),$$

and by (P4) along with the lemma,

$$\sum_{\substack{-1 < q < 1 \\ q \in \mathbb{Q}}} \lambda(\Omega + q) \leq \lambda((-1, 2)) = 3.$$

But by (P3),  $\lambda(\Omega + q) = \lambda(\Omega)$ , and so

$$\sum_{\substack{-1 < q < 1 \\ q \in \mathbb{Q}}} \lambda(\Omega) \leq \lambda((-1, 2)) = 3,$$

which is only possible if  $\lambda(\Omega) = 0$ , since there are infinitely many rationals between  $-1$  and  $1$ . Hence,  $\lambda(\Omega) = 0$ .

**We now show:**

$$(0, 1) \subseteq \bigsqcup_{\substack{-1 < q < 1 \\ q \in \mathbb{Q}}} (\Omega + q).$$

Suppose  $x \in (0, 1)$ . Let  $x_0 = [x] \cap \Omega$ . Then,  $x \sim x_0$ , i.e.,  $x - x_0 = q \in \mathbb{Q}$ . Now,  $x_0 \in (0, 1)$  (for  $x_0 \in \Omega \subseteq (0, 1)$ ), and  $x \in (0, 1)$ , which implies that  $q = x - x_0 \in (-1, 1)$ . Thus,  $x = x_0 + q \in \Omega + q$  for some rational  $q$  such that  $-1 < q < 1$ . Hence,  $x \in \bigsqcup_{\substack{-1 < q < 1 \\ q \in \mathbb{Q}}} (\Omega + q)$ .

By the lemma,

$$\lambda((0, 1)) \leq \lambda \left( \bigsqcup_{\substack{-1 < q < 1 \\ q \in \mathbb{Q}}} (\Omega + q) \right),$$

i.e.,

$$1 \leq \sum_{\substack{-1 < q < 1 \\ q \in \mathbb{Q}}} \lambda(\Omega + q),$$

which is a contradiction, for the sum on the right is 0.

**QED**